

Clun

*JavaScript and TypeScript toolkit * pure Common Lisp * <https://clun.sh>*

Reply to Attn of: REL/CLUN

TO: Maintainers and Users

FROM: Project Engineering

SUBJECT: Release memo - Clun 0.3.0 (shared-memory workers and zero-stubs close)

REF: Issue #338, Issue #339, PR #340, PR #341, Epic #177

Bottom line: Clun 0.3.0 is the minor train that lands shared-memory multithreading (SharedArrayBuffer, Atomics, real worker_threads) and closes the zero-stubs inventory so exported Node and runtime surfaces are not hollow no-ops. Capability matrix remains 30 Yes / 0 Partial / 0 No. Until this candidate publishes, the installable boundary stays v0.2.2.

1. Why

Full-port law forbids soft Yes and hollow exports. Workers without real shared memory, and Node surfaces that ignore arguments and return undefined, break the exceed-Bun contract. Issues #338 and #339 complete that unit for the 0.3.0 minor release.

2. What changed

- SharedArrayBuffer, Atomics, and real worker_threads (Issue #338 / PR #340)
- WASI path_open, fd_read, and path_filestat_get against host files (pure CL)
- process.stdout and process.stderr end plus writableEnded stream semantics
- MessagePort start records _started; Node-compatible auto-start retained
- module.register(string) records _registeredSpecifiers
- Final destub fixture tests/js/node/destub-zero-final.js
- Residuals inventory closed: ZERO_PRODUCT_HOLLOWS (docs/design/zero-stubs-residuals.md)
- WebSocket wss: via pure-tls worker path (related surface train)
- Source version staged 0.3.0; release ledger candidate tracks Issue #339

3. SemVer disposition

Impact: minor. Rationale: backward-compatible capability completion (shared-memory multithreading and residual destubs). Version: 0.3.0. Tag: v0.3.0. Previous published boundary: 0.2.2 / v0.2.2. Pre-1.0 note: minor versions on the 0.x train may still include breaking changes per project SemVer policy.

4. Intentional design no-ops (not hollows)

- MessagePort start: records `_started`; ports auto-start as in Node
- crypto FIPS: always reports 0; pure-CL crypto is not FIPS-validated
- `v8.startupSnapshot.isBuildingSnapshot`: honest false
- `os.setPriority` for foreign pids: `SystemError` (no forbidden kernel niceness API)

5. Gates completed on the close unit

5.1 make build

5.2 make purity (0 violations)

5.3 make test (JS and TS fixtures green; parachute Failed: 0)

5.4 Mechanical ignore-args hollow scan: zero product hollows

6. Install note (publication state)

While the candidate remains unpublished, `curl -fsSL https://clun.sh/install | sh` reinstalls the verified boundary v0.2.2. After v0.3.0 assets publish and the ledger advances, the same command and `clun --update target 0.3.0`. Always use `man clun` after install; `man` content must match live CLI help.

6.1 Current verified install

```
curl -fsSL https://clun.sh/install | sh
```

6.2 After 0.3.0 publishes, update in place

```
clun --update
```

7. Documents for this ship

- Operator SOP: `docs/sop-clun-ops.pdf`
- This release memo: `docs/memo-release-0.3.0.pdf`
- `CHANGELOG.md` Unreleased / 0.3.0 section
- `README.md` and <https://clun.sh> Documents links
- Capability matrix: `compat/` (30 Yes)

8. Progressive depth (not soft Yes)

Depth limits such as full HPACK peer interop, full CDP, and V8-precise coverage maps remain progressive quality work. They do not justify hollow exports. Exported APIs perform real work rather than ignore to undefined.

Project Engineering
Clun release

Date: 2026-07-27

cc: ['https://clun.sh', 'github.com/theesfeld/clun']